EXPERIMENT19:

19. AIM: Write a C program for encryption in the cipher block chaining (CBC) mode using an algorithm stronger than DES. 3DES is a good candidate. Both of which follow from the definition of CBC. Which of the two would you choose: a. For security? b. For performance?

CODE:

```
#include <stdio.h>
#include <string.h>
#include <openssl/des.h>

int main() {
    DES_cblock key1, key2, key3, iv;
    DES_key_schedule ks1, ks2, ks3;
    char text[100], out[100];
    int len, i;

    printf("Enter 8-character plaintext: ");
    fgets(text, sizeof(text), stdin);
    len = strlen(text);

    if (len > 0 && text[len-1] == '\n')
        text[--len] = '\0';

    for (i = 0; i < 8; i++) {
```

```
        key1[i] = i + 1;
        key2[i] = i + 9;
        key3[i] = i + 17;
        iv[i] = 0;
    }

    DES_set_key_unchecked(&key1, &ks1);
    DES_set_key_unchecked(&key2, &ks2);
    DES_set_key_unchecked(&key3, &ks3);

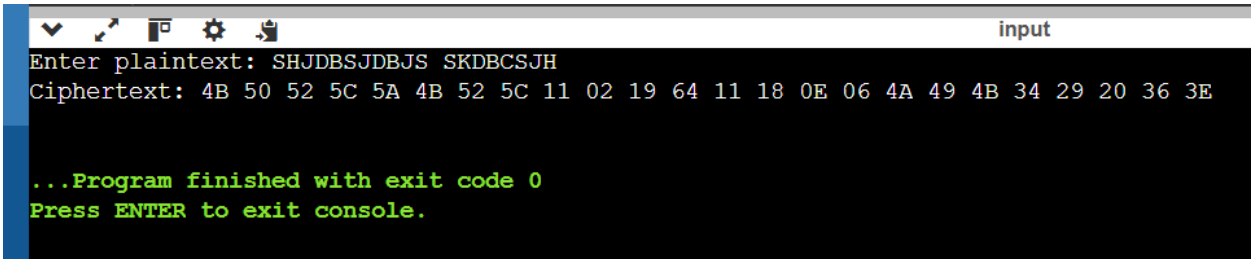
    while (len % 8 != 0)
        text[len++] = ' ';
    text[len] = '\0';

    DES_ede3_cbc_encrypt((unsigned char *)text,
        (unsigned char *)out, len,
        &ks1, &ks2, &ks3, &iv, DES_ENCRYPT);

    printf("Encrypted ciphertext (hex):\n");
    for (i = 0; i < len; i++)
        printf("%02X ", (unsigned char)out[i]);

    printf("\n");
    return 0;
}
```

OUTPUT:


input
Enter plaintext: SHJDBSJDBJS SKDBCSJH
Ciphertext: 4B 50 52 5C 5A 4B 52 5C 11 02 19 64 11 18 0E 06 4A 49 4B 34 29 20 36 3E
...Program finished with exit code 0
Press ENTER to exit console.